S571 S783 S2648 S8754 S669 S461 Ssym Q3421 K3822 K2369 PTMUL PTMUL-D mean of 12

GeoDDG-3D
DDGemb
ESM-MSR (devel) combined
ESM-MSR combined
ESM-MSR (devel) WT head
ESM-MSR WT head
MutateEverything st1
MutateEverything st2
DDMut
SPURS
ThermoMPNN + ThermoMPNN-D
ThermoMPNN
ESM3ΔG
ESM3ΔG
SPURS multi
ProteinMPNN (zero-shot)
ESM3-medium (Forge)
ESM3-small (Forge)
ProSST (zero-shot)
ESM3-small-open, σ = 0 (zero-shot)
ESM3-large (Forge)
IFUM, WT backbone*
IFUM, ESMFold*
ESMtherm*

0.25 0.50 | 0.25 0.50 0.75 | 0.25 0.50 0.75 | 0.25 0.50 0.75 | 0.25 0.50 | 0.25 0.50 0.75 | 0.25 0.50 0.75 | 0.25 0.50 0.75 | 0.25 0.50 0.75 | 0.25 0.50 0.75 | 0.0 0.5 | 0.0 0.5 | 0.25 0.50 0.75

Spearman ρ (95% protein-level bootstrap); red ring = documented training overlap with the set. * trained on Megascale (Tsuboyama) data: not held out (grey rows)